SQL MERGE Statement

The **MERGE** statement in SQL Server is used to perform **INSERT**, **UPDATE**, and **DELETE** operations in a single statement. It compares data from a **source table** with a **target table** based on a matching condition. Depending on whether rows match or do not match, SQL Server can update existing rows, insert new rows, or delete rows that no longer exist in the source.

The MERGE statement is commonly used in **data synchronization**, **data warehousing (ETL)**, and **keeping two tables consistent**. Instead of writing separate `INSERT`, `UPDATE`, and `DELETE` statements, MERGE combines all these operations into one, making the code easier to manage and maintain.

Syntax

```
MERGE TargetTableAS TUSING SourceTableAS SON T.ID= S.IDWHEN MATCHEDTHENUPDATESET T.Name= S.NameWHENNOT MATCHEDBY TARGETTHENINSERT (ID, Name)VALUES (S.ID, S.Name)WHENNOT MATCHEDBY SOURCETHENDELETE;
```

Example

Suppose we have two tables:

Employees

ID	Name	Salary
1	Ahmed	5000
2	Sara	6000

NewEmployees

ID	Name	Salary
2	Sara	6500
3	Ali	4500

The following MERGE statement synchronizes the `Employees` table with the `NewEmployees` table:

```
MERGE EmployeesAS TUSING NewEmployeesAS SON T.ID= S.IDWHEN MATCHEDTHENUPDATESET
        T.Name= S.Name,
        T.Salary= S.SalaryWHENNOT MATCHEDBY TARGETTHENINSERT (ID, Name, Salary)VALUES (S.
ID, S.Name, S.Salary)WHENNOT MATCHEDBY SOURCETHENDELETE;
```

Result

After executing the statement:

ID	Name	Salary
2	Sara	6500
3	Ali	4500

- Employee **Sara** was updated because the ID existed in both tables.
- Employee **Ali** was inserted because the ID existed only in the source table.
- Employee **Ahmed** was deleted because the record no longer existed in the source table.

Advantages of MERGE

- Combines **INSERT**, **UPDATE**, and **DELETE** into a single statement.
- Improves code readability and reduces duplication.
- Efficient for synchronizing large datasets.
- Commonly used in ETL processes and data warehouse applications.
- Reduces the need for multiple SQL statements and transactions.

Conclusion

The SQL **MERGE** statement is a powerful feature that simplifies data synchronization between two tables. By combining multiple operations into one statement, it makes database maintenance more efficient and easier to understand. It is especially useful in real-world applications where data from different sources must be kept consistent.